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SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

CHAPTER 3:Practices for Requirements Elicitation Part 1

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Contents

3.1 Sources for requirements – stakeholders, documents, other systems
3.2 Requirements categorization using Kano Model

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Objectives

- Apply techniques to identify and extract requirements from multiple sources such as stakeholders, documents, and other systems in a given project scenario
- Analyze the relationships and influence among different stakeholders to determine their relevance and impact on system requirements.
- Categorize system features using the Kano Model to distinguish between basic, performance, and excitement requirements.

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Practices for Requirements Elaboration

- The **quality of requirements** directly influences the quality of the development output.
- Without well-defined requirements, it is unlikely that the resulting system will meet user needs or be effective.
- Therefore, it is essential to **elaborate requirements professionally** to ensure clarity and usefulness.

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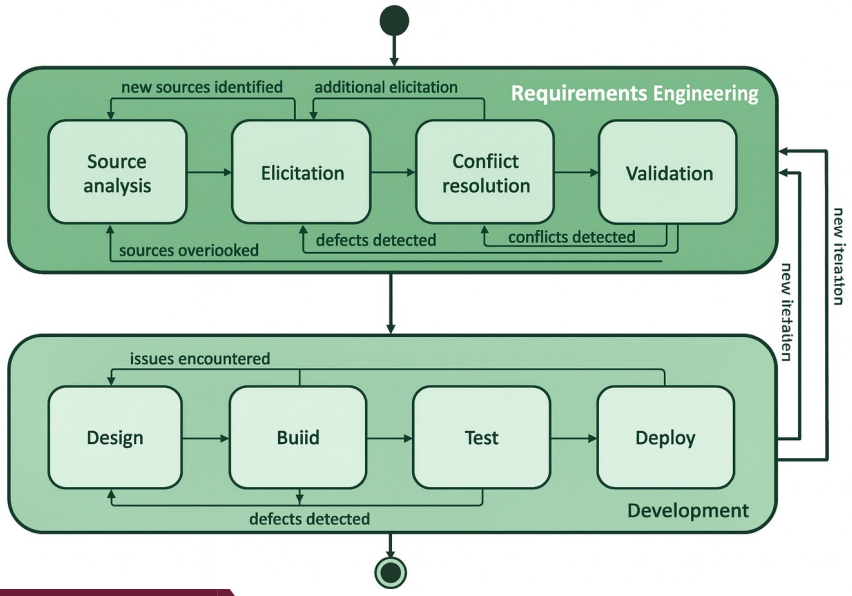
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3.1 SOURCES OF REQUIREMENTS

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Tasks in Requirements Engineering

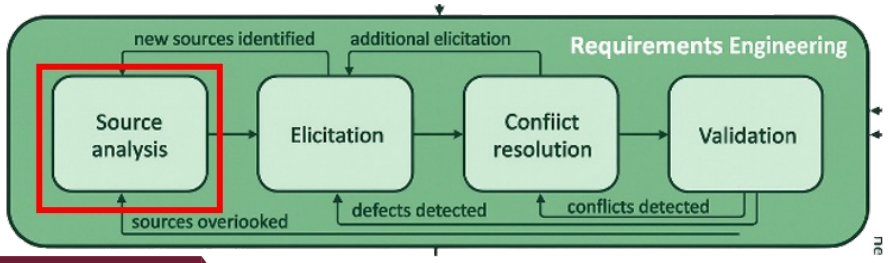


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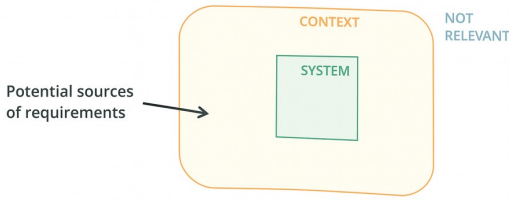
Sources for Requirements

- Identification of requirements sources is an iterative and recursive that requires constant reconsideration.
- The quality and completeness of requirements depend greatly on the requirements depend greatly on the requirements sources involved.
- Missing a relevant source will lead to an incomplete understanding of the requirements or to incomplete requirements.



Context and Sources

- Understand the context of the system to be developed in a certain application domain.
- Necessary to identify the relevant requirements sources.



Types of Requirements Sources



Stakeholder

- A stakeholder is either a person or an organization with potential interest in the desired system.
- Has knowledge about several context objects and their relationships
- Each stakeholder typically has specifi goals and requiremenets for the desired system.
- Examples:
  - User, employer, client, manager…
  - Operator, developer, architect, tester



Documents

- Contain important information that can provide requirements.
- Examples:
  - Universal documents : laws, norms, standards
  - Domain-or-organization-specific documents : concepts, professional articles, printed papers, incidents/error reports of legacy systems



Systems-in-Operation

- •Give the stakeholders a chance to try out the systems, so:
  - Gain impression of the current system
  - Request for extensions or changes
- Examples:
  - Existing or legacy systems
  - External system
  - Competitor systems

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3.1 SOURCES OF REQUIREMENTS: STAKEHOLDER

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Requirements Sources: Stakeholder

- **Definition:** A stakeholder is any individual or organization that influences a system's requirements or is affected by that system.
- **Role:** They are the primary sources of requirements for a system.
- **Importance:** If a key stakeholder is not identified, it can severely impact the quality of the final requirements.
- **Risks of Late Discovery:** Identifying stakeholders too late (or overlooking them entirely) can result in costly modifications or, in the worst case, a system that fails to meet real needs.
- **Best Practice:** Begin the process of identifying stakeholders systematically and as early as possible.

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Typical Stakeholder Roles

- Users/end users
- Sponsors
- Managers
- Developers
- Authorities
- Customers

- People or organizations that are impacted by a system are (indirect) stakeholders

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Organization as Stakeholders

- It is useful to view stakeholders as real people, not just abstract entities.

- When identifying an organization as a stakeholder, consider:
  - Who is the responsible person for this organization?
  - Who can act as the main contact point?
  - Who officially represents this organization within our company?

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Organization as Stakeholders

- **Example:**
  - The **government** may be a stakeholder if certain laws apply.
  - Identify a representative of the government who can provide the necessary requirements.
  - It is not practical to name a high-level figure such as the Prime Minister.
  - A more effective choice would be someone closer to the matter, e.g., the **head of the internal legal department**.

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Stakeholder Analysis

- **Key Question for Identification:**

When determining if a person or organization should be considered a stakeholder, ask:

"Does a relevant relationship exist between the person/organization and the system?"

Note:

There is no universal or standardized method for identifying stakeholders.

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Potential sources for identifying relevant stakeholder and stakeholder roles are:



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Minimum Information on Stakeholder List

- Gather essential details about each stakeholder and maintain them in a stakeholder list.
- Keep this list updated throughout the requirements engineering (RE) process.
- At a minimum, the list should include:
  - **Name**
  - **Role or function**
  - **Contact and personal information**
  - **Availability** (time and location during project progress)
  - **Importance/Relevance** of the stakeholder
  - **Expertise** (area and level of knowledge)
  - **Goals and interests** related to the project

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Stakeholder List Example

Name	Function / Role	Contact	Temporal Availability	Spatial Availability	Relevance / Importance	Expertise	Goals
Joe Smith	Head of University Library Council	smith@c.lib.org	10% commited to project	15 New St Putrajaya	High	Library Business Process	Operational cost savings of 25%
Virigina Wolf	UTM Library Clerk	v.wolf@utm.my	Friday full day	PSZ Library UTM	Medium	Knowledge about customers	Better usability of new system
Alex Wright	UTM Sudent Key User	a.wright@utm.my	Mon, Wed, Fri afternoon	Fac of Computing UTM	High		Improved management of borrowed items
Sue Emmet	Chief Editor	sueem@bho.com	On request 1 week in advanced	Book House Publishing, Malaysia	Medium	Publishing Processes	Easy management of Journal subscription
John Doe	System Architect	johndoe@lif.utm.my	80% committed to project	Lib Soft, UTM	Medium	Security Expert	Customer data is secure
Yue Wen	Tester	yue@testfit.my	80% committed to project	Testit Kuala Lumpur	Low	Automated UI Testing	All UI test are automated

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Stakeholder Relationship Management

- Issues with stakeholders may occur when:
  - The rights and responsibilities of stakeholders and requirements engineers are unclear.
  - The needs of stakeholders are not adequately addressed.
- To avoid misunderstandings, establish a stakeholder agreement or stakeholder contract.
- Endorsement and support from senior management can greatly increase the likelihood of success.

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A Special Stakeholder: The User

- In systems with a user interface, end-users form a key stakeholder group, especially important for requirements engineers.
- User acceptance is critical to the overall success of the project.
- Users require special attention throughout the requirements engineering process.
- **For projects with large user bases, users should be grouped into categories (user groups) to simplify requirements elicitation**.
- One effective method for representing large user groups is through **personas**.

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Large Number of Users

- Systems may have thousands or even millions of users.
- It is not feasible to tailor requirements to meet every individual's needs.
- Instead, users should be grouped into categories based on behavioral similarities, making each group distinct from others.
- Typically, dividing users into **three to seven groups** works best.
- Each group should be treated as a unique source of requirements.

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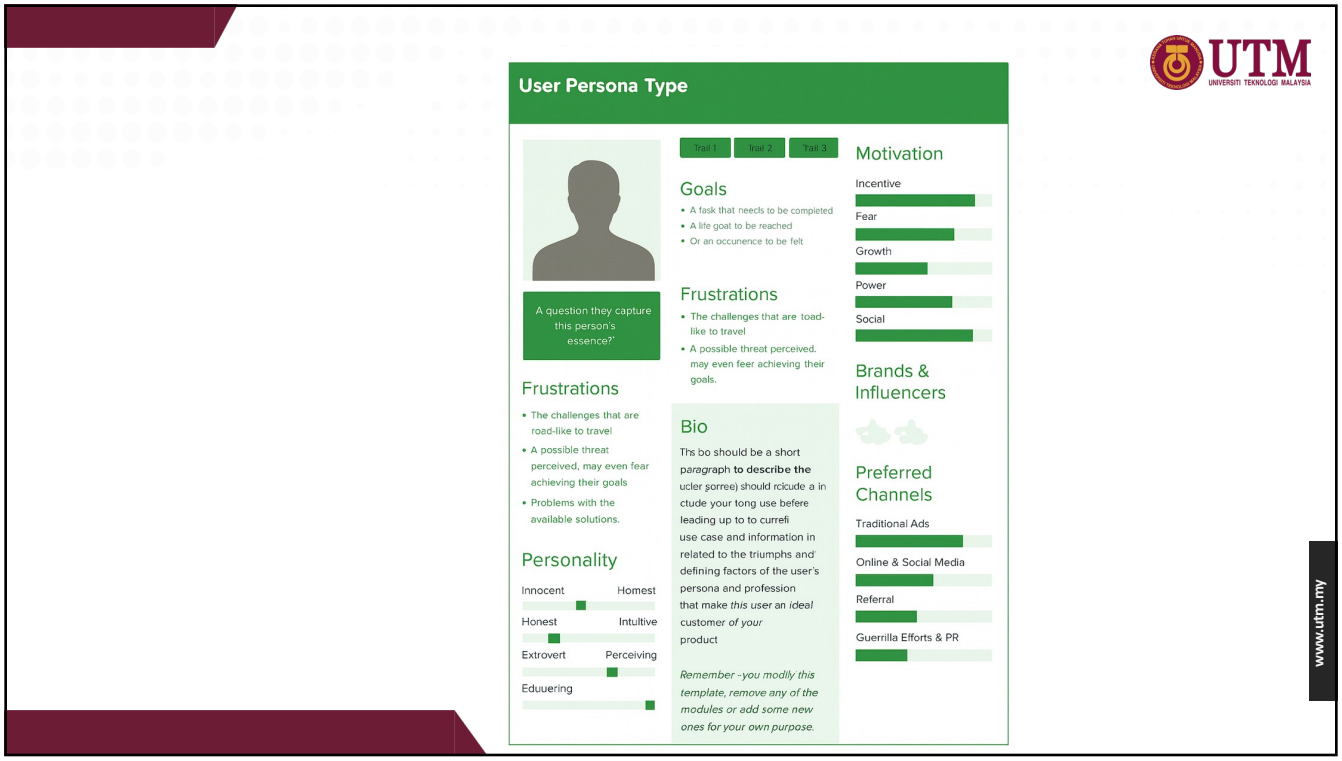
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User Groups and Personas

- A useful technique is the creation of **personas**.
- **Personas** are fictional characters that represent typical users or groups of users who share common goals, needs, behaviors, or attitudes.
- They help to better understand and design for user groups.
- Personas are usually developed based on **user research** or **market research**.

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User Groups and Personas - Example

User Persona – UTM Library System

Name	Aisyah Rahman
Role / Title	Undergraduate Student – Faculty of Computing
Age	22
Gender	Female
Education Level	Bachelor's Degree (Year 3)
Location	Kolej Tun Fatimah, Universiti Teknologi Malaysia
Tech Proficiency	Intermediate (comfortable with digital platforms)

Background:

Aisyah is a full-time Computer Science student who frequently uses the UTM Library for both academic research and personal reading. She borrows reference books, accesses online journals, and occasionally books study rooms for group discussions. Aisyah prefers digital solutions and values systems that are intuitive, quick, and mobile-friendly.

Goals and Motivations:

- Find and borrow books easily without spending too much time searching.
- Renew books online and avoid overdue fines.
- Access e-journals and digital resources anytime, anywhere.
- Receive notifications for due dates and newly added books.
- Have a seamless experience between the library website and the UTM student portal.

Pain Points / Challenges:

- Difficult to find the exact book location within shelves.
- Sometimes forgets to return books on time, resulting in fines.
- Library system feels outdated and slow on mobile.
- Hard to discover related materials or recommended readings automatically.

Needs and Expectations:

- Mobile-friendly library system interface.
- Smart search with filters (author, topic, availability).
- Automatic due-date reminders via email or WhatsApp.
- Personalized book recommendations based on borrowing history.
- Integration with UTM Smart Card for borrowing and payments.

Quote:

"I just want a fast and smart library system that helps me find what I need without wasting time."

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3.1 SOURCES OF REQUIREMENTS: DOCUMENTS

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Documents

- Documentation serves as a valuable source for identifying requirements.
  - Various types of documents may be relevant:
  - Company, domain, and project-related documents
  - Product and process descriptions
  - Legal and regulatory documentation
- Especially useful for capturing requirements that are often overlooked or not explicitly mentioned by stakeholders, such as:
  - Constraints defined in standards, company policies, or legal/regulatory guidelines
  - Detailed specifications in procedures and work instructions
  - Innovative ideas found in competitor marketing materials

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Types of Documents

General Binding Documents

- This type of documents subsume documents defined by standardization bodies, governments, special interest groups, trade or professional organizations or the like.
- Examples:
  - laws and standards

Organization-specific documents

- This type of documents are defined within an organization involved in the development or use of the desired system.
- Examples:
  - guidelines, product strategies, HCI guideline, organization procedure, organization business process

Product/System-specific documents

- Documents describing any kind of development artefacts (eg. requirements, code, architecture, test cases) of a predecessor system or similar system.
- Examples:
  - Change request document, error reports, test documentation, requirements specifications

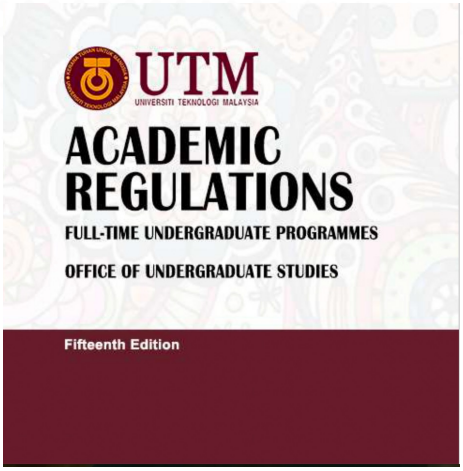
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Documents Example





General Binded Document

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Documents

- Reviewing documentation provides better understanding of the **context** in which a system is to be developed.
- Exploring similar solutions used in other companies or domains can:
- Offer insights into common problems and goals
  - Spark creativity and new ideas
  - Suggest feasible directions for the current project

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Working with Documents

- Maintain a **document list** (similar to a stakeholder list) to keep track of all documents.
- Store documents in a **central, indexed repository** with unique IDs for easy reference.
- Always note **dates and version numbers** to avoid using outdated documents.
- Review and update documents regularly to ensure any changes are reflected in the requirements.

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Working with Documents

- Whenever possible, rely on the **final versions** of documents.
- Keep **older versions archived**, as they may provide insight into how a system and its requirements have evolved.
- Establish a **clear and reliable document management process** from the beginning of the project to ensure accuracy and consistency.

3.1 SOURCES OF REQUIREMENTS: SYSTEMS-IN-OPERATION

Types of Systems

Predecessor Systems

- Predecessor (legacy) systems typically have a similar purpose as the system to be developed.
- They are still in use within an organization (or have been in use) and are typically (partial) replaced by the new system.

Systems of competitors

- Analysing systems competing with the system to be developed is essential to not oversee important requirements and to ensure the system offers differentiating features for its users.

Systems from other domains

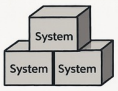
- Systems developed for another domain might offer properties, use innovative technology or offer unique features relevant for the desired system.
- Such systems could be valuable sources for innovative requirements for the desired system.

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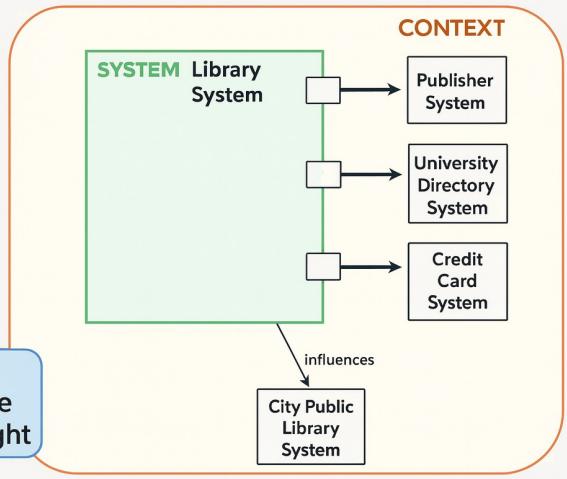
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Other Systems



- Different types of systems to consider:
  - Internal and external
  - Similar and interfacing

? What are these categories for the systems on the right

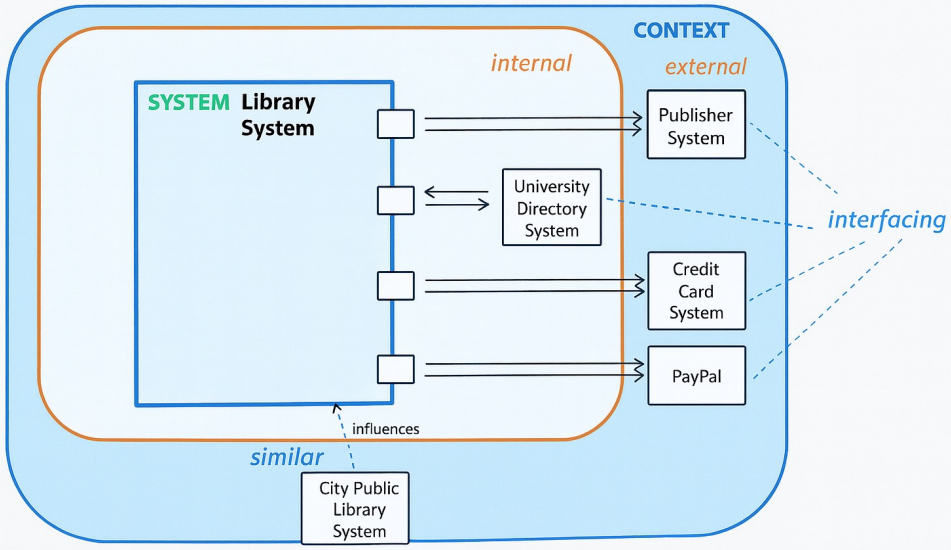


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Sytems in Context of Library System



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3.2 REQUIREMENTS CATEGORIZATION USING KANO MODEL

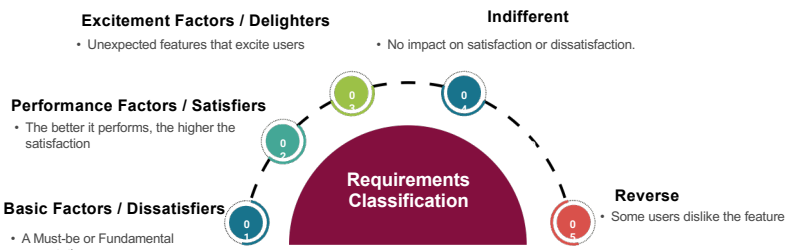
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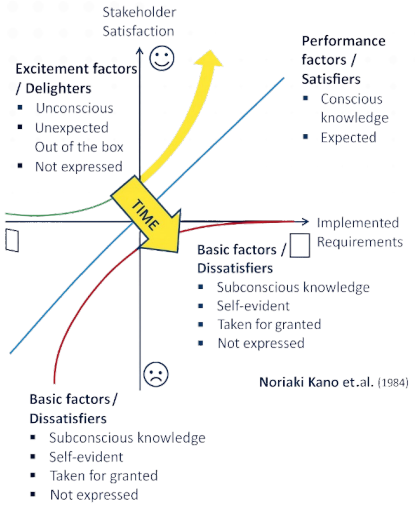
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Kano Model

A customer satisfaction model developed by Noriaki Kano to classify product or system requirements based on how they affect user satisfaction.



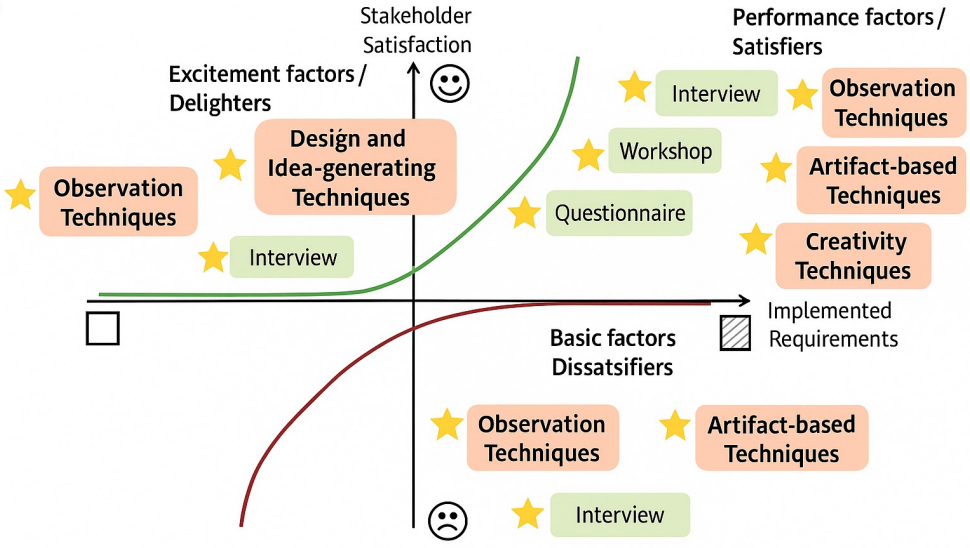


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Kano Model and Elicitation Techniques



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Kano Model and Elicitation Techniques



Interview can find basic factors when using techniques such as "typical work day" where the interviewer guides the interviewee through a past experience in every detail.

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Kano Model and Elicitation Techniques



- **Interview** can uncover excitement factors by asking creativity questions, e.g., a *"magic question"* ("Imagine a djinn appears and grants you three wishes for the new system, what would they be?").
- **Observation** can reveal excitement factors when a stakeholder does something unexpected or out of the box, which could indicate an excitement factor.

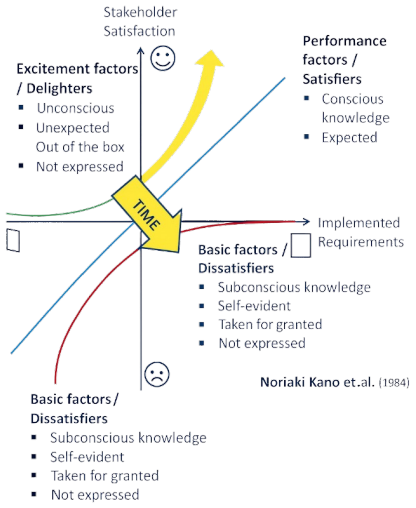
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Example of Library System

Factors	Requirements / Features
Must Be / Basic	
Performance	
Excitement	
Indifferent	
Reverse	



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